DAY 3 – Linux Fundamentals & Log Analysis

Objective

The objective of this session is to develop a strong foundation in **Linux system operations** and introduce **Security Operations Center (SOC)–oriented log analysis techniques**. This hands-on exercise is designed to simulate real-world scenarios where security analysts monitor, investigate, and respond to system activities using logs.

By the end of this session, the learner is expected to:

- Demonstrate proficiency in essential Linux commands
- Navigate system directories and manage files effectively
- Analyze system and authentication logs
- Identify anomalies and potential security threats
- Apply analytical thinking to real-world cybersecurity scenarios

LEVEL 1 – BASIC

Task 1: Linux Command Familiarization

Understanding basic Linux commands is essential for interacting with systems in a SOC environment.

Command	Description	Use Case
ls	Lists files and directories	Viewing contents of a directory
pwd	Prints current working directory	Identifying current location
cd	Changes directory	Navigating file system
cat	Displays file content	Reading logs and files

Outcomes:

These commands form the foundation for **system navigation, log access, and file handling**, which are critical for any cybersecurity professional.

Task 2: Directory Exploration

Objective:

To explore the Linux file system and locate important system logs.

Steps:

1. Open terminal
2. Navigate to /var/log
   - *cd /var/log*
3. List all files
   - *ls*

Key Log Files Observed:

- syslog – General system activity logs
- auth.log – Authentication and login records
- kern.log – Kernel-related logs
- dpkg.log – Package installation logs

Outcomes:

The /var/log directory is a central location for system logs, making it a primary source for **incident detection and forensic analysis**.

Task 3: File Creation

Steps:

1. Create a file named test.txt
   - *touch test.txt*
2. Add any text inside
   - *echo "Sample log analysis practice" > test.txt*
3. View the file
   - *cat test.txt*

Outcomes:

- File creation and modification
- Content verification
- Understanding file handling in Linux

LEVEL 2 – INTERMEDIATE

Task 4: Log Reading

Command:

- *cat /var/log/syslog*

What type of logs do you see?

System logs typically include:

- Service start/stop events
- Scheduled tasks (cron jobs)
- System errors and warnings
- Background process activities

Write 3 log entries :

Apr 16 10:15:32 systemd[1]: Started Session 1
Apr 16 10:16:01 CRON[1234]: Job executed
Apr 16 10:17:45 kernel: Device initialized

Analysis:

These logs help in identifying:

- System behavior patterns
- Unexpected failures
- Unauthorized processes

Task 5: Authentication Log

Command:

- *cat /var/log/auth.log*

login-related entries such as:

- Login attempts (successful/failed)
- SSH access logs
- User authentication events

Example Entries:

- Accepted password for user from 192.168.1.5
- Failed password for invalid user admin from 192.168.1.10

Outcomes:

Authentication logs are critical for detecting:

- Unauthorized access attempts
- Brute force attacks
- Suspicious login behavior

Task 6: Keyword Search

Command:

- *grep "Failed" /var/log/auth.log*

Purpose:

How many failed attempts?

- Counts failed login attempts

What does it indicate?

- Identifies suspicious access attempts

Outcomes:

A high number of failed login attempts may indicate:

- Password guessing attacks
- Automated brute force attempts

LEVEL 3 – ADVANCED

Task 7: Suspicious Activity Detection

Indicators of Suspicion:

- Multiple failed login attempts in a short duration
- Access attempts from unknown IP addresses
- Login attempts outside normal working hours

Example:

Repeated failures from a single IP may indicate an attack pattern.

Task 8: Pattern Identification

Key Patterns to Identify:

- Repeated login attempts by the same user
- Continuous access attempts from the same IP
- Sequential failures followed by success

Example Pattern:

Failed login from 192.168.1.10 repeated multiple times

Conclusion:

Such patterns strongly suggest a **brute force or credential stuffing attack**.

Task 9: Create Your Own Fake Log

Sample Logs:

Failed password for user admin from 192.168.1.10
Failed password for user admin from 192.168.1.10
Accepted password for user admin from 192.168.1.10
Failed password for user root from 192.168.1.20
Failed password for user test from 192.168.1.30

Purpose:

Simulating logs helps in:

- Understanding log structures
- Practicing threat identification
- Preparing for real SOC scenarios

Task 10: CRITICAL THINKING ANALYSIS

Scenario:

"If a system shows 50 failed login attempts within 1 minute"

Professional Analysis:

This behavior strongly indicates a **brute force attack**, where an attacker uses automated tools to attempt multiple password combinations rapidly. Such high-frequency attempts are abnormal for legitimate users and suggest malicious intent. If successful, this could lead to unauthorized access and potential system compromise. Immediate mitigation steps should include blocking the source IP, enabling account lockout mechanisms, and enforcing multi-factor authentication (MFA). Continuous monitoring and alerting systems (SIEM) play a crucial role in detecting such anomalies in real time.

Task 11: BONUS ANALYSIS

- **Successful vs Failed Login**

Type	Description
Successful Login	Valid credentials used; access granted
Failed Login	Invalid credentials; access denied

Task 12: Why are logs important in cybersecurity?

Logs are a critical component of cybersecurity because they:

- Provide visibility into system and user activities
- Enable detection of security incidents
- Support forensic investigations
- Help in compliance and auditing
- Assist SOC teams in real-time monitoring and response